

IN INDIA'S FIRST HINDI PROGRAMMING LANGUAGE

लिपि

L I P I

भारत की प्रोग्रामिंग भाषा

Write code the way India thinks — in Devanagari script & Hindi keywords

15

PHASES

258

TESTS PASS

100%

PURE RUST

∞

LOVE FOR भारत

⚙ Built with **Rust** · Bytecode VM · WASM Playground · Roman Input · TCO · Enums

● ● ● namaste.swami

। भारत की पहली हिंदी प्रोग्रामिंग भाषा । विधि नमस्ते(नाम) : फल "नमस्ते, " + नाम + "! IN" बताओ
नमस्ते("दुनिया") → नमस्ते, दुनिया! IN

2 What is LIPI & Why does it exist?

What is LIPI?

LIPI (लिपि) is India's first full-featured programming language written entirely in **Devanagari script** with **Hindi & Sanskrit keywords**. Every keyword, every builtin, every error message speaks in the language 600 million Indians read and write every day.

From **विधि** (function) to **यदि/अन्यथा** (if/else) to **जब तक** (while) — the syntax reads like natural Hindi. Variables use Devanagari names, numbers can be written in Hindi digits (१, २, ३...) and even in Indian units (५ लाख = 500,000).

"If a child in Bhopal can learn algebra in Hindi, why should she need to switch to English to write her first program?"

— THE CORE PHILOSOPHY OF LIPI

Why was LIPI built?



Language Barrier in CS Education

700M Hindi speakers are locked out of programming because every language is in English.



Cultural Identity in Technology

India contributed zero, Fibonacci via Virahanka, trigonometry via Āryabhaṭa — yet our tech speaks English.



Proof of Concept

Demonstrate that a production-quality, bytecode-compiled language can be built in pure Rust with non-Latin syntax.



Bridge for New Programmers

Teach programming concepts to Hindi-medium students without the cognitive load of learning English simultaneously.

Development Timeline

PHASE 1-4 · CORE

Lexer, parser, AST, arithmetic, variables, conditionals, WASM playground

PHASE 5-6 · DATA STRUCTURES

Lists, dicts, classes, inheritance, object methods

PHASE 7-9 · I/O & ERRORS

While loops, break/continue, REPL, try/catch, string interpolation

PHASE 10-11 · FUNCTIONS

Lambdas, true closures, HOFs (map/filter/reduce), iterable for-loops

PHASE 12-13 · LANGUAGE+

Varargs, ternary, bitwise ops, dict iteration, boolean keywords, globals

PHASE 14 · भारत MATH

Ancient Hindu math (Virahanka/Āryabhaṭa/Brahmagupta), Indian number formatting, स्वरूप()

PHASE 15 · COMPLETENESS

Enums (विकल्प), pattern matching (मिलाओ), TCO, triple-quote strings, ₹ formatting

PHASE 16 · NEXT

Module system, generators (उत्पन्न), named format placeholders

3 Features & Code Examples



Devanagari Syntax

Full Unicode support — keywords, variables, strings, even number literals in Hindi digits



Bytecode VM

AST compiled to LVM opcodes, not tree-walked. Fast execution with stack-based VM



WASM Playground

Runs in any browser — compile to WebAssembly with wasm-pack, zero install needed



Roman Input

Write LIPI phonetically — type "batao", "yadi", "vidhi" and it auto-translitrates



Tail-Call Opt.

TCO for deep recursion — tail calls reuse the current frame, no stack overflow



Enums & Match

Algebraic data types with payload, exhaustive pattern matching (मिलाओ/विकल्प)



भारत Stdlib

Aadhaar validation (Verhoeff), UPI, GST, EMI, ancient Hindu math functions



Indian Numbers

5 लाख, 2 करोड़ literals; ₹12,34,567 formatting; Devanagari digit support



258 Tests

Comprehensive stress test — every feature, every edge case, all green ✓

INDIAN NUMBERS & FORMATTING



आयात भारत.संख्या राशि है 5 लाख | 500,000 | बड़ा है २ करोड़
| 20,000,000 | बताओ स्वरूप("{:₹}", 1234567) →
₹12,34,567 बताओ लाख_में(750000) → 7.50 लाख बताओ
रुपये_में(राशि) → ₹5,00,000

ENUMS & PATTERN MATCHING



विकल्प मौसम: वर्षा(मिमी) धूप बादल म है मौसम.वर्षा(42) मिलाओ
म: वर्षा(x): बताओ "बारिश: " + x + " mm" धूप: बताओ "☀"
धूप है!" अन्यथा: बताओ "बादल" → बारिश: 42 mm

ANCIENT HINDU MATH (भारत.गणित) & CLOSURES

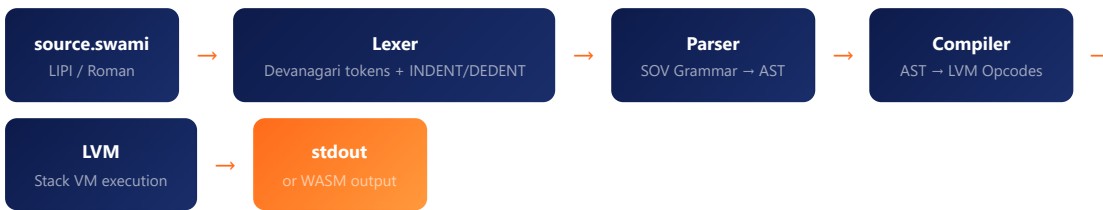


आयात भारत.गणित | Virahanka Fibonacci (600 CE) | बताओ
विरहांक(10) → 55 | Bhāskara II Combinations (1150 CE) |
बताओ संयोजन(10, 3) → 120 | Āryabhaṭa Trigonometry (499
CE) | बताओ ज्या(पाई / 6) → 0.5

| True Closures | विधि गुणक(n): फल लाम्बा(x): x * n दोगुना
है गुणक(2) तिगुना है गुणक(3) बताओ दोगुना(7) → 14 बताओ
तिगुना(5) → 15 | HOF pipeline | बताओ मोड़ो(मानचित्र(
छानो([1..10], लाम्बा(x): x%2==0), दोगुना), 0, लाम्बा(a,b):
a+b)

4 Architecture, Tech Stack & What's Next

⚙️ Execution Pipeline



60+

OPCODES

8

SOURCE FILES

5K+

LINES OF RUST

4

STDLIB MODULES

0

DEPENDENCIES*



Rust

Core language, zero deps



WebAssembly

wasm-pack browser port



LVM

Custom bytecode VM



.swami / .libc

Source & binary formats

📅 Roadmap

✅ COMPLETE

- 📖 Devanagari syntax
- ⚡ Bytecode VM
- 🔗 Enums + match
- 📦 TCO
- 🌐 WASM
- 📄 Roman input
- 🏠 भारत stdlib

🚧 PHASE 16

- 📦 Module system
- 🏷️ Named placeholders
- 🔧 Generators (उत्पन्न)
- 📖 Better REPL

🔮 FUTURE

- 💻 VS Code extension
- 📖 Learning platform
- 🌐 Other Indian scripts
- 📦 Package registry

🌟 VISION

- 📖 Teach CS in Hindi
- 📖 School curriculum
- 📖 Hindi CS books
- 👥 Community LIPI

जय हिंद। जय LIPI। IN

If you believe technology should be accessible to every Indian regardless of language, share this, star the repo, and let's build India's coding future — in Hindi.



Built in Pure Rust



Open Source



DM for collaboration

*except wasm-bindgen for the browser playground · All 258 tests pass · Works on Windows, Linux, macOS